



COS80023 Big Data

Credit Task 10 Clustering vs. Classifiers

Overview

With the help of given R commands, run a simple k-means clustering algorithm. Compare its results with the results produced by random forest (RF), which you investigated in Pass task 8.

Purpose

Learn to use R for clustering. Learn the difference between clustering and classification, and the difference between supervised and unsupervised learning.

Task

Go through the steps described below. Answer the questions in a separate document.

Time

This task should be completed in your 10th tutorial or the week after and submitted to Canvas for feedback. It should be discussed and signed off in tutorial 11 or 12.

This task should take no more than 2 hours to complete (excluding introductory videos).

Resources

- <https://www.youtube.com/watch?v=4b5d3muPQmA>
- Code listing from Pass task 8 (below)
- Any other material you find useful in explaining the results
- You can use genAI for your research or additional clarification

Feedback

Demonstrate your steps and discuss your answers with the tutorial instructor.

Next

Compile your portfolio or keep working on your D/HD project.

Credit Task 10 — Submission Details and Assessment Criteria

Write down the questions and answers in a text or Word document, convert to pdf and upload to Canvas. Your tutor will mark the submission on line. If the submission is not marked as '1', it is considered as incomplete and must be resubmitted.

Task 10

Code for Loading Data

```
library("caret")
```

(This loads the caret library which has functions that we need. If you need to know what a function does, you can ask for help using `?`, e.g. `?traincontrol()`).

```
iris.data <- read.csv(file.choose())
```

```
View(iris.data)
```

```
iris.data$species <- as.factor(iris.data$species)
```

```
set.seed(32984)
```

```
indexes <- createDataPartition(iris.data$species, times = 1,  
                               p = 0.7, list = FALSE)
```

```
train <- iris.data[indexes,]
```

```
test <- iris.data[-indexes,]
```

```
trainctrl <- trainControl(method = "repeatedcv", number = 10,  
                          repeats = 3)
```

Random Forest Classification

To remind yourself of classification, apply the Random Forest algorithm to the dataset as you did in Pass task 8.

```
rfmodel <- train(species ~., data=train, method="ranger",  
                 trControl=trainctrl,
```

```
                 preProcess = c("center", "scale"), tuneLength=10)
```

```
rfresult <- predict(rfmodel, newdata=test)
```

Clustering using k-means

k-means is a very simple and intuitive clustering algorithm. Before you start, watch the video introduction which illustrates the k-means process:

<https://www.youtube.com/watch?v=4b5d3muPQmA>

The function to perform k-means clustering in R is simply:

```
kmeans(<data>, k)
```

Question 1. What variable should you enter for each of the parameters? Why?

Question 2. For clustering, should we use the train, test or complete dataset? Why?

Once you have investigated and made informed decisions, run the `kmeans()` function and assign the result to `kmeansresult`.

Show the `kmeansresult` on the screen. You will see a listing of components at the end. You can view each component by typing `kmeansresult$<component>`.

`kmeansresult$cluster` shows the cluster number (assigned by `kmeans()`) for each entry.

Comparing the results

Compare the results of kmeans clustering and random forest.

First, you have to find the results of both kmeans and random forest for a matching number of rows.

You can combine the results in a dataframe df. To convert data into a data frame you can use

```
df <- as.data.frame(<mycolumns>)
```

To combine columns into a single table, you can use

```
df <- (df, <othercolumns>)
```

Question 3. When you compare the cluster assignment of kmeans with the species prediction of random forest, do these agree? Can you tell which cluster is which species? Discuss in about five sentences.

You can find out more using a scatterplot.

```
ggplot(data=iris.data, aes(x=sepal_width, y=petal_length,  
color=species)) + geom_point() + theme_minimal()
```

shows two of the three variables of the iris data set, with the colouring according to the dependent variable.

Add the kmeans cluster number as an additional column to the data set. Choose this column for the colouring. Make scatterplots of different combinations of the independent variables. Take a screenshot of the most illustrative combination(s) and put it into your answer document, then use it to answer

Question 4. Can you see why kmeans() assigned the clusters as it did? State your observations.